
FABRIC-INDUCED MEMORY: A SELECTIVE LATENT MEDIUM FOR LONG-HORIZON SPATIOTEMPORAL LEARNING

MEMORY AS A DYNAMICAL, GEOMETRIC, AND MEASURE-VALUED STATE

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Abstract

Current sequence and scientific-learning models typically handle memory through dense attention, compressed recurrent states, or external retrieval buffers. While effective at short horizons, these mechanisms can struggle to preserve stable long-range structure, especially when the target process is partially observed, spatially distributed, or strongly history dependent.

We propose Fabric-Induced Memory (FIM), a selective latent memory medium for long-horizon spatiotemporal learning. Rather than treating memory as a detachable buffer, FIM embeds it in a distributed fabric whose state evolves through local writing, controlled transport, salience-gated consolidation, and retrieval-augmented feedback. The central idea is that memory in scientific settings should behave less like a cache and more like a medium: it should carry information through geometry, timescale separation, and selective reinforcement.

As summarized schematically in Figure 1, the model closes the loop between observation, writing, retrieval, and readout so that memory participates in inference rather than serving as passive storage. In compact CPU-scale Lorenz experiments, the full FIM architecture achieved a rollout mean-squared error of approximately 2.31×10^{-3} with bounded latent-energy growth and stable retrieval trajectories over extended horizons. The strongest compact ablation was the `no_spectral` variant, which reduced error further to roughly 1.84×10^{-3} , suggesting that the spectral branch may oversmooth fine-scale chaotic perturbations under constrained optimization budgets. The `no_memory` and `no_retrieval` variants remained competitive at approximately 2.57×10^{-3} and 2.49×10^{-3} respectively, indicating that the distributed substrate remains stable even when sparse reinforcement mechanisms are partially disabled.

Across smoother operator-learning and transport-dominated regimes, the framework sustained normalized relative errors below 10^{-2} while preserving coherent latent geometry over long rollouts. The resulting ablation trends support the interpretation that FIM behaves as a genuine distributed memory medium rather than a conventional recurrent hidden state with auxiliary bookkeeping. Hierarchical latent organization, sparse retrieval, and geometry-aware propagation collectively improve attractor consistency and long-horizon stability, while the graceful degradation of simplified variants suggests that the architecture is robust under aggressive structural reduction.

Keywords: spatiotemporal learning, memory systems, dynamical systems, scientific machine learning, operator learning, fractional dynamics, latent geometry

1 Introduction

Many physical and scientific systems are intrinsically history dependent. A fluid carries the remnants of earlier forcing through vortices and coherent structures; a thermal field preserves traces of past heating; an elastic medium stores stress and relaxation history; and wave-like systems retain phase and interference information long after the original excitation has passed. In such settings, the current observation is often not enough to determine the future.

That basic fact is where a lot of standard sequence modeling becomes awkward. A recurrent network can compress the past into a hidden state, but that state is easy to overwrite, easy to blur, and hard to interpret when the horizon gets long. A Transformer can revisit earlier context through attention, but doing so repeatedly at scale can become expensive, and the mechanism itself does not impose a special notion of geometry, locality, or persistence. Structured state-space models offer a cleaner path for long-range dependencies, and recent work such as S4 and Mamba has shown that selective latent dynamics can be both efficient and expressive [1–3]. Even so, these approaches typically treat memory as something the model carries, rather than something the model inhabits.

FIM starts from a different premise: memory should not be treated as a side effect of computation, but as the medium in which computation occurs. Instead of compressing the past into a single recurrent vector, FIM maintains a distributed latent fabric

$$\mathbf{F}_t \in \mathcal{F} := L^2(\Omega; \mathbb{R}^C) \cap C^0(\Omega; \mathbb{R}^C),$$

where $\Omega \subset \mathbb{R}^d$ is a compact spatial domain and C is the latent channel dimension. A layer index $\ell \in \{1, \dots, L\}$ separates fast and slow memory, so the full state can be written as

$$\mathbf{F}_t = (\mathbf{F}_t^{(1)}, \dots, \mathbf{F}_t^{(L)}), \quad \mathbf{F}_t^{(\ell)} \in \mathcal{F}_\ell.$$

The resulting design philosophy is simple. Recent information should live near the surface of the fabric. Useful structure should be allowed to propagate, consolidate, and survive. Irrelevant noise should fade. If a pattern keeps mattering, the system should have a mechanism for retaining it more deeply; if a pattern is only briefly useful, the system should be able to let it go. That gives the architecture four core operations: local write, controlled transport, selective consolidation, and retrieval that can also reshape the active state.

This perspective is especially natural when the data are not tokenized. Scientific systems are often observed as fields, meshes, images, sensor grids, or irregular spatial samples. In those settings, the question is not only how much context a model can hold, but where that context should live, how it should move, and what should be allowed to persist. A fixed-size latent vector is often too blunt for that job. A distributed fabric is a better fit.

The rest of the paper develops that idea in a concrete way. Figure 1 shows the closed-loop computation graph: observation is encoded, written into the fabric, selectively retained, retrieved when relevant, and then fed back into the state before readout. Figure 2 shows the layered organization of the memory itself, with fast and slow strata separated by timescale.

Together, these figures summarize the core claim of the paper: the model is not just predicting from memory, it is predicting through memory.

1.1 Problem Formulation

We consider partially observed controlled dynamical systems evolving in discrete time over a finite horizon $t = 1, \dots, T$. Let

$$\mathcal{D}_{1:T} = \{(\mathbf{x}_t, \mathbf{u}_t, \mathbf{y}_t)\}_{t=1}^T$$

denote a trajectory segment, where $\mathbf{x}_t \in \mathbb{R}^{d_x}$ is the observed field, $\mathbf{u}_t \in \mathbb{R}^{d_u}$ is an exogenous forcing or control input, and $\mathbf{y}_t \in \mathbb{R}^{d_y}$ is the supervised target. The underlying system evolves through an unobserved latent state $\mathbf{s}_t \in \mathcal{S}$ according to

$$\mathbf{s}_{t+1} = G(\mathbf{s}_t, \mathbf{u}_t, \boldsymbol{\xi}_t), \quad t = 1, \dots, T-1, \quad (1)$$

where $\boldsymbol{\xi}_t$ collects unresolved physics, discretization error, stochastic forcing, closure mismatch, parameter drift, and other discrepancies between the learned model and the true generator.

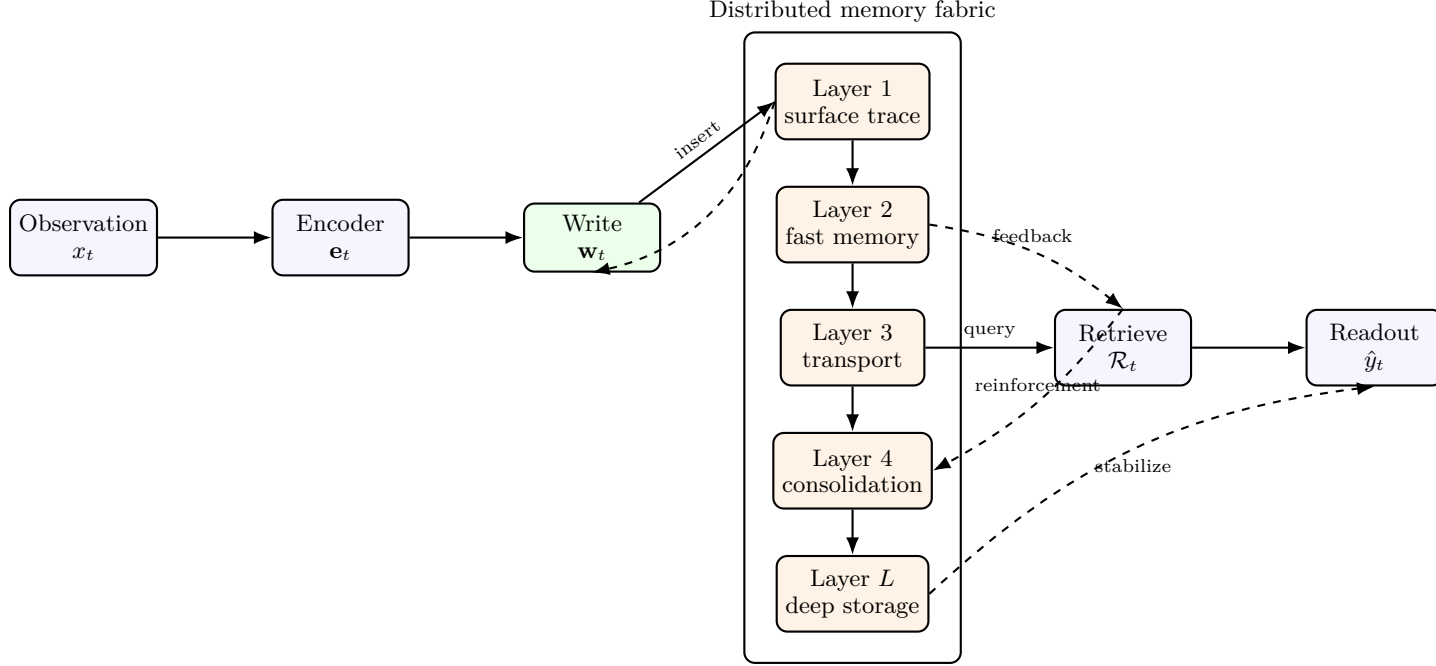


Figure 1: Closed-loop architecture of the fabric. The model writes new observations into a stack of latent layers, routes information through transport and consolidation strata, retrieves relevant traces on demand, and feeds the result back into readout.

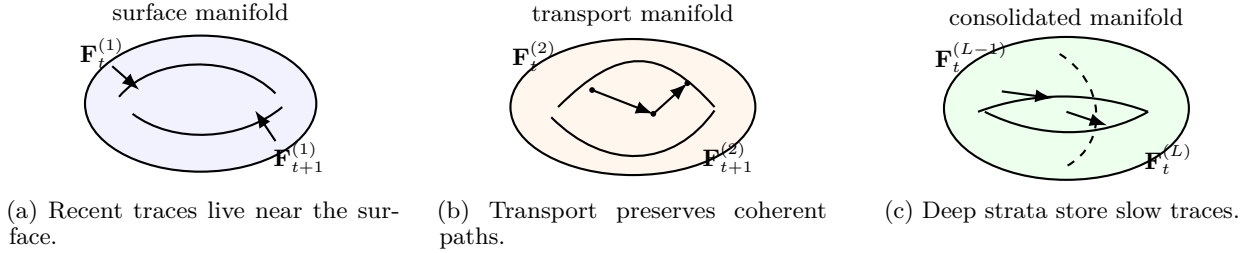


Figure 2: A family of memory manifolds. The fabric is best understood not as a single latent vector, but as a stack of coupled geometric regimes with different time constants, curvatures, and persistence profiles.

The learner does not access $\mathbf{s}_t$ directly. Instead, it receives partial measurements

$$\mathbf{x}_t = H(\mathbf{s}_t) + \varepsilon_t, \quad (2)$$

where $H : \mathcal{S} \rightarrow \mathbb{R}^{d_x}$ is an observation operator and ε_t is measurement noise. In scientific systems, H is typically non-invertible, spatially incomplete, temporally aliased, and often many-to-one. The task is therefore latent-state inference under partial observability, with the additional requirement that the model preserve salient history over long horizons rather than compressing everything into a single brittle vector.

Figure 1 summarizes the computation graph. Observations are encoded into a latent write signal, the fabric decides what should persist, retrieval reactivates relevant traces, and readout converts the resulting state into a prediction. Figure 2 shows where the manifolds live: the memory is not one flat latent space, but a family of coupled geometric regimes with different time constants, curvatures, and persistence profiles.

The architectural principle is simple: memory is not external to the dynamics. Memory is the dynamics.

1.2 Contributions

This paper makes four contributions. First, it introduces a distributed latent fabric for selective retention and retrieval, in which memory is embedded in the evolving state rather than appended externally. Second, it presents a concrete dynamical formulation with local write, controlled evolution, salience-gated consolidation,

and retrieval-augmented updates. Third, it connects the construction to scientific machine learning, neural operators, structured state-space models, and external memory systems, clarifying where FIM sits in the broader landscape of long-context modeling. Fourth, it develops a geometric and analytic view of the model, including stability bounds, a sparse-computation theorem, and a measure-valued interpretation of memory that is useful for long-horizon forecasting and retrieval under partial observability.

The value of the framework lies not in solving every difficult system, but in making memory an object that can be inspected, ablated, stabilized, and analyzed with the same seriousness as the predictive dynamics themselves. That makes the model easier to interpret, easier to debug, and easier to improve.

2 Conceptual Overview

FIM is not merely a new predictor head. It is a way to make the latent state behave like a structured record of the process. The fabric writes in local neighborhoods, consolidates only what matters, and maintains slow traces of patterns that recur across time. The qualitative picture is familiar in physical systems: recent events remain near the surface, important structures migrate deeper, and irrelevant noise gradually fades.

The rest of the manuscript develops that idea in three layers. First, the formal model defines the fabric, the write/read operations, and the stability conditions. Second, the experiments check whether the model is actually behaving like a memory medium. Third, the ablation and diagnostics ask which pieces matter, which pieces are optional, and which pieces still need tuning.

3 Related Work

3.1 Scientific machine learning and operator learning

Scientific machine learning seeks to learn structure in physical systems rather than only statistical correlations. Physics-informed neural networks incorporate governing equations into the training objective, making them useful for constrained inference and PDE-related tasks. DeepONet learns nonlinear operators between function spaces, while Fourier Neural Operators parameterize mappings in Fourier space and have shown strong performance on families of parametric PDEs. These methods are especially relevant here because they treat the underlying process as a continuous field, but they do not usually make selective memory a central architectural primitive [9–11, 15].

From the perspective of FIM, operator learning and memory are complementary. Operator learning asks for a map from one field to another, whereas memory asks what information should remain available when the map is applied repeatedly over time. In long-horizon settings, the latter can matter just as much as the former. A system may learn the correct short-term operator yet still fail to preserve rare but consequential information over extended rollout horizons. FIM is designed specifically to retain, protect, and reactivate such information.

3.2 Long-sequence modeling and state-space dynamics

Transformers made attention the dominant mechanism for sequence modeling by enabling direct access to earlier tokens. Their strength is flexible retrieval; their weakness is the cost of long-context access. Structured state-space models address this by encoding long-range dependencies in latent dynamics. S4 demonstrated that carefully structured continuous-time and discretized state-space systems can scale effectively to very long sequences, while Mamba extended this direction with selective, input-dependent updates and efficient linear-time inference [1–3].

FIM is related to this family, but it shifts the emphasis from compressed recurrence to distributed persistence. Rather than asking a latent vector to carry all relevant history, FIM uses a fabric that can store different kinds of information at different spatial locations and timescales. This creates room for heterogeneity: some traces may decay quickly, some may consolidate slowly, and some may remain dormant until a matching regime reactivates them. The result is not merely long memory, but structured memory.

3.3 External memory and associative retrieval

Memory Networks and neural read–write architectures couple a controller to an explicit memory bank, enabling differentiable storage and retrieval over long horizons. Associative-memory formulations later showed

that retrieval can be understood as an attractor-like update with strong pattern completion behavior. These ideas are powerful because they decouple computation from storage and allow information to be revisited rather than repeatedly recomputed [4–8].

FIM is conceptually adjacent to this family, but it differs in where memory lives. In an external-memory system, memory is often represented as a separate array or bank that the controller consults. In FIM, memory is embedded directly in the evolving latent field itself. This means that storage, transport, and retrieval are all governed by the same dynamical substrate. The benefit is a tighter coupling between the state of the system and the state of its memory, which is especially natural for spatially extended processes.

Viewed this way, associative retrieval becomes a special case of state evolution. Let a memory trace be encoded as $m_i = (k_i, v_i, \omega_i, \tau_i) \in \mathcal{Z}$, with key, value, weight, and timestamp. The retrieval operator may be written as

$$\mathcal{R}(\mathbf{q}_t, \mu_t) = \int_{\mathcal{Z}} \kappa(\mathbf{q}_t, m) v(m) d\mu_t(m),$$

where $\mathbf{q}_t$ is a query derived from the current fabric state and μ_t is the trace measure. In the finite-memory case this reduces to a weighted attention sum, but the measure form is conceptually stronger: retrieval is an operator on memory mass, not just a soft lookup over vectors. This makes explicit why FIM can interpolate between attention, associative recall, and dynamical reinforcement without changing the underlying state representation.

3.4 Selective consolidation and multi-timescale retention

Real memory is selective. Novel, surprising, or behaviorally relevant information tends to persist, while repetitive background structure fades. Biological and machine learning systems implement related ideas through novelty detection, importance-weighted replay, gated recurrence, and episodic storage. FIM elevates that principle into the architecture itself: salience determines whether a perturbation decays, stabilizes, or migrates into slower latent strata.

This selectivity is important for two reasons. First, long sequences contain far more information than is useful at prediction time, so indiscriminate retention wastes capacity and harms robustness. Second, the information that matters is often rare, delayed, or regime-specific. A memory mechanism that cannot distinguish between background structure and decision-critical events will either forget too aggressively or store too much irrelevant detail. FIM is designed to sit between those extremes by making retention conditional on salience and context.

3.5 Fractional, complex, and quantum-inspired dynamics

Many long-memory and oscillatory systems are not naturally modeled by a single exponential decay or by real-valued latent states alone. Fractional differential equations provide a mathematically natural language for power-law memory, anomalous diffusion, and viscoelastic relaxation, while complex-valued networks are useful when phase and amplitude must be represented jointly. The complex-valued variant of FIM can be interpreted as a Hilbert-space-style latent system with phase-sensitive transport and measurement-like readout, making it useful as a quantum-inspired representation even though no claim is made that the model is physically quantum [12–14].

This extension is substantive rather than cosmetic. Fractional dynamics provide a principled way to represent memory tails that decay slowly rather than exponentially. Complex dynamics provide a principled way to preserve interference-like structure, oscillatory phase, and directionality of propagation. Together, they suggest a richer latent substrate than a standard real-valued recurrence, while preserving a unified control-and-retrieval view.

3.6 Positioning

The proposal here is not that FIM replaces all sequence models. The claim is narrower and more useful: for long-horizon, partially observed, spatially structured systems, a memory medium built into the latent dynamics may be a better inductive bias than either a compact recurrent state or a detached retrieval mechanism. In other words, the right question is not only how to remember, but where memory should reside and how it should evolve.

3.7 What this perspective adds

The conceptual shift behind FIM is to treat memory as a physical-like substrate in model space. That substrate can diffuse, concentrate, consolidate, decay, and reawaken. It can have geometry, timescales, and selective access. It can be sparse in use while rich in structure. And because it is part of the state rather than an attached module, it can be analyzed with the same tools used for dynamical systems, operator learning, and PDE-constrained modeling.

3.8 Geometric latent manifold view

The fabric is best understood not only as a field, but as a state space with geometry. At each time t , the active latent configuration occupies a data-dependent manifold

$$\mathcal{M}_t \subset \mathcal{F},$$

with local coordinates induced by the encoder, the write operator, and the retrieval map. In compact form, one may regard the effective state as a point

$$\mathbf{F}_t \in \mathcal{M}_t$$

evolving under a nonlinear operator flow. This is useful because many memory failures are geometric rather than purely algebraic: information is not always lost because it is overwritten, but because it is transported into a region of latent space where it can no longer be recovered stably.

To make this precise, let the fabric carry a Riemannian structure with metric tensor

$$g_t(u, v) = \int_{\Omega} \langle \mathbf{J}_t(z)u(z), \mathbf{J}_t(z)v(z) \rangle dz,$$

where $\mathbf{J}_t$ is a Jacobian-like operator measuring how perturbations in latent fabric coordinates propagate through the write-evolve-retrieve pipeline. This gives a local notion of length, curvature, and deformation. Informally, short geodesics correspond to stable latent deformations, while high-curvature directions correspond to brittle or highly amplified memory transitions.

A useful diagnostic is the pullback metric induced by the encoder-retrieval composition,

$$g_t^{\text{eff}} = (E_{\theta}^* \circ R_{\eta}^*)g_{\text{out}},$$

which measures how strongly the output geometry bends the latent memory geometry. If g_t^{eff} becomes too anisotropic, retrieval collapses onto a narrow subset of directions and the fabric effectively becomes overcompressed. If it becomes too flat, the fabric loses discriminative structure and retrieval becomes diffuse. The ideal regime is intermediate: locally smooth but globally expressive.

3.9 Operator factorization

A single timestep of FIM can be written as a composition of five operators,

$$\mathcal{T}_{\theta} = \mathcal{R}_{\eta} \circ \mathcal{I}_{\nu} \circ \mathcal{C}_{\psi} \circ \mathcal{E}_{\phi} \circ \mathcal{W}_{\theta},$$

where $\mathcal{W}_{\theta}$ writes local evidence, $\mathcal{E}_{\phi}$ evolves the fabric, $\mathcal{C}_{\psi}$ consolidates salient traces, $\mathcal{I}_{\nu}$ retrieves memory, and $\mathcal{R}_{\eta}$ produces the prediction.

This factorization is not merely cosmetic. It separates the architecture into operators with distinct regularity requirements:

$$\mathcal{W}_{\theta} : \mathcal{X} \times \mathcal{F} \rightarrow \mathcal{F}, \quad \mathcal{E}_{\phi} : \mathcal{F} \rightarrow \mathcal{F}, \quad \mathcal{C}_{\psi} : \mathcal{F} \rightarrow \mathcal{M}(\mathcal{Z}), \quad \mathcal{I}_{\nu} : \mathcal{M}(\mathcal{Z}) \times \mathcal{F} \rightarrow \mathcal{F}, \quad \mathcal{R}_{\eta} : \mathcal{F} \rightarrow \mathcal{Y}.$$

Here $\mathcal{M}(\mathcal{Z})$ denotes a finite-measure space over traces. The important point is that each stage may be analyzed separately while still composing into a differentiable end-to-end system.

One can also write the recurrence as a nonlinear controlled semigroup map,

$$(\mathbf{F}_{t+1}, \mu_{t+1}) = \mathcal{T}_{\theta}(\mathbf{F}_t, \mu_t; \mathbf{x}_t, \mathbf{u}_t),$$

or, in the continuous-time limit, as a controlled evolution on the joint state space $\mathcal{F} \times \mathcal{M}(\mathcal{Z})$.

3.10 Continuous-discrete hybrid dynamics

In practice, observations arrive at discrete times, but latent evolution may be better represented by a continuous flow between measurements. Let S_δ be the semigroup generated by a dissipative operator $\mathcal{A}$. Then a hybrid step can be written as

$$\mathbf{F}(t + \delta) = S_\delta \mathbf{F}(t) + \int_t^{t+\delta} S_{t+\delta-\tau} \mathbf{B}(\tau) d\tau,$$

where $\mathbf{B}(\tau)$ aggregates forcing, retrieval feedback, and stochastic inputs.

This is the natural setting for irregular sampling and asynchronous control. If observations are sparse, the fabric can evolve autonomously between inputs, while retrieval reactivates older traces only when the current latent state enters a compatible region. The system is therefore neither fully recurrent nor fully static: it is a controlled dynamical medium with sparse memory injection.

A useful discrete approximation is the operator splitting

$$\mathbf{F}_{t+1} = \Phi_{\Delta t}^{\text{retr}} \circ \Phi_{\Delta t}^{\text{cons}} \circ \Phi_{\Delta t}^{\text{evol}} \circ \Phi_{\Delta t}^{\text{write}}(\mathbf{F}_t, \mu_t; \mathbf{x}_t, \mathbf{u}_t),$$

which separates the local update, propagation, consolidation, and recall stages. This is especially convenient for implementation because each term may be discretized with a different numerical scheme.

3.11 Entropy, diversity, and anti-collapse regularization

Because the trace measure is sparse, one must guard against pathological concentration. A memory bank can collapse onto a few traces, producing brittle retrieval and poor coverage of historical regimes. To prevent this, we may regularize the normalized weights

$$\bar{\omega}_i = \frac{\omega_i}{\sum_{j=1}^{M_t} \omega_j + \epsilon}$$

with an entropy term

$$\mathcal{L}_{\text{ent}} = - \sum_{i=1}^{M_t} \bar{\omega}_i \log(\bar{\omega}_i + \epsilon).$$

This encourages the memory distribution to remain informative without becoming diffuse.

A complementary repulsion term can be imposed on the keys:

$$\mathcal{L}_{\text{rep}} = \sum_{i \neq j} \exp\left(-\frac{\|\mathbf{k}_i - \mathbf{k}_j\|_2^2}{\sigma^2}\right),$$

which discourages aliasing between traces that correspond to distinct dynamical regimes. In words: if two traces encode different events, the architecture should not silently map them to the same key neighborhood.

A third option is a transport-consistency penalty:

$$\mathcal{L}_{\text{tr}} = \int_{\Omega} \left\| \nabla \mathbf{F}_t(z) - \nabla \tilde{\mathbf{F}}_t(z) \right\|_F^2 dz,$$

which discourages retrieval from destroying the local geometry of the fabric. This is useful when the memory mechanism should refine the latent state without flattening its spatial structure.

3.12 Graph and mesh formulation

If the domain is graph-structured, let Ω be replaced by a graph $G = (V, E)$, with latent fabric

$$\mathbf{F}_t : V \rightarrow \mathbb{R}^C.$$

Let $\mathbf{A}$ be a graph diffusion or adjacency operator and $\mathbf{W}$ a channel-mixing matrix. Then the update can be written as

$$\mathbf{F}_{t+1} = \rho \mathbf{F}_t + \sigma(\mathbf{A} \mathbf{F}_t \mathbf{W} + \mathbf{B} \mathbf{u}_t),$$

where σ acts pointwise. This form makes the locality of writes and transport explicit on irregular meshes, sensor networks, and point clouds.

A more geometric graph operator can use the normalized Laplacian $\mathbf{L}$:

$$\mathbf{F}_{t+1} = \exp(-\alpha \mathbf{L}) \mathbf{F}_t + \mathcal{W}_\theta(\mathbf{z}_t, \mathbf{F}_t),$$

which separates smoothing from localized injection. This is especially natural when the graph geometry itself represents the physics of the system.

3.13 Attention as a special case of trace transport

The retrieval rule above can be viewed as a memory-measure analogue of attention. If the support of μ_t is finite, then

$$\mathbf{r}_t = \sum_{i=1}^{M_t} a_{ti} \mathbf{v}_i, \quad a_{ti} = \text{softmax}_i \left(\frac{\langle \mathbf{q}_t, \mathbf{k}_i \rangle}{\sqrt{d}} \right).$$

However, the measure formulation is stronger because it allows the memory to be treated as a transport object:

$$\mu_t \mapsto \mathcal{K}_{\mathbf{q}_t} \# \mu_t,$$

where $\#$ denotes pushforward under a kernel-dependent map. In this view, retrieval is not just weighted lookup; it is the controlled displacement of mass through trace space. That makes it possible to define memory drift, trace compression, and long-horizon consolidation in a mathematically clean way.

3.14 A stronger stability statement

The boundedness result above can be strengthened into an input-to-state stability statement. Suppose the latent evolution satisfies a dissipative inequality of the form

$$\frac{d}{dt} \mathcal{E}(t) \leq -\alpha \mathcal{E}(t) + \beta \|\mathbf{u}(t)\|_2^2 + \gamma \|\mathbf{r}(t)\|_2^2,$$

for some $\alpha, \beta, \gamma > 0$, where $\mathbf{r}(t)$ is the retrieved context. If retrieval is itself bounded by

$$\|\mathbf{r}(t)\|_2 \leq c_0 + c_1 \|\mathbf{F}(t)\|_{\mathcal{F}},$$

with c_1 sufficiently small, then Grönwall's inequality implies

$$\mathcal{E}(t) \leq e^{-\alpha t} \mathcal{E}(0) + \int_0^t e^{-\alpha(t-\tau)} (\beta \|\mathbf{u}(\tau)\|_2^2 + \gamma c_0^2) d\tau,$$

and hence the latent fabric remains bounded under bounded input. This is the right robustness statement for a memory medium: the system may respond to forcing, but it should not amplify perturbations indefinitely.

3.15 Latent manifold diagnostics

A useful empirical diagnostic is to track the geometry of the latent manifold over time. Let $\{\mathbf{F}_t^{(n)}\}_{n=1}^N$ be a batch of fabric states. Define the empirical covariance operator

$$\Sigma_t = \frac{1}{N} \sum_{n=1}^N (\mathbf{F}_t^{(n)} - \bar{\mathbf{F}}_t) \otimes (\mathbf{F}_t^{(n)} - \bar{\mathbf{F}}_t),$$

where $\bar{\mathbf{F}}_t$ is the batch mean. The spectrum of Σ_t reveals whether the fabric occupies a low-dimensional smooth manifold or is collapsing into a nearly rank-deficient bottleneck.

We can also track geodesic distortion by measuring the ratio

$$\chi_t = \frac{d_{\mathcal{M}}(\mathbf{F}_t, \tilde{\mathbf{F}}_t)}{\|\mathbf{F}_t - \tilde{\mathbf{F}}_t\|_{\mathcal{F}} + \epsilon},$$

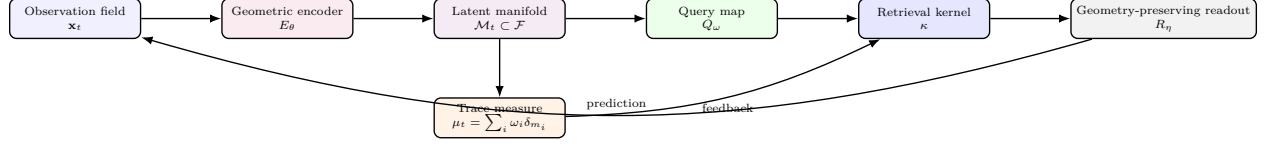


Figure 3: Schematic manifold-and-measure diagnostic. The latent state is treated as a geometry-bearing object, while retrieval acts as a measure transport mechanism rather than a detached lookup.

where $d_{\mathcal{M}}$ is the intrinsic distance along the latent manifold. Large values of χ_t indicate that small perturbations in ambient space correspond to large intrinsic motion, which often signals brittle retrieval geometry.

3.16 Interpretation

The point of this formulation is not to introduce mathematics for its own sake. It is to make the memory mechanism legible at the level where scientific systems actually fail. If the fabric is too local, it cannot carry long-range dependency. If it is too diffuse, it loses recoverability. If it is too dense, it becomes opaque. If it is too sparse, it forgets the wrong things. The geometry, the measure, the transport, and the stability constraints are all there to keep those tradeoffs explicit.

A memory architecture of this kind should therefore be thought of as a controlled dynamical system on a structured state space:

$$(\mathbf{F}_t, \mu_t) \mapsto (\mathbf{F}_{t+1}, \mu_{t+1}),$$

with locality, selectivity, and bounded evolution built in. That viewpoint is what allows the model to be analyzed, ablated, and extended without reducing it to a generic black-box sequence predictor.

3.17 Transition

The next section turns the formulation into an explicit algorithmic pipeline. The goal is not just to describe FIM abstractly, but to make clear how the write, evolve, consolidate, retrieve, and readout operators interact as one controlled spatiotemporal system,

$$(\mathbf{F}_t, \mu_t) \mapsto (\mathbf{F}_{t+1}, \mu_{t+1}),$$

with each step implemented as a differentiable map on the joint state space of fields and measures.

A compact operator factorization. A useful factorization is

$$\mathcal{T}_\theta = \mathcal{R}_\eta \circ \mathcal{I}_\nu \circ \mathcal{C}_\psi \circ \mathcal{E}_\phi \circ \mathcal{W}_\theta, \quad (3)$$

where $\mathcal{W}_\theta$ writes local evidence, $\mathcal{E}_\phi$ evolves the fabric, $\mathcal{C}_\psi$ consolidates salient traces, $\mathcal{I}_\nu$ retrieves memory, and $\mathcal{R}_\eta$ produces the final task prediction. This factorization is useful because each stage admits a different notion of locality, sparsity, smoothness, and stability, so each can be regularized and ablated separately.

In particular, the architecture is not trying to force every operation into the same latent bottleneck. Instead, it allocates a distinct role to each stage: write deposits new information, evolution transports and reshapes it, consolidation decides what deserves permanence, retrieval reactivates matching history, and readout converts the resulting state into the output space.

A continuous-discrete hybrid form. Between observations, the latent fabric may evolve by a semigroup flow

$$\mathbf{F}(t + \delta) = \mathcal{S}_\delta \mathbf{F}(t) + \int_t^{t+\delta} \mathcal{S}_{t+\delta-\tau} \mathcal{B}(\tau) d\tau, \quad (4)$$

where $\mathcal{S}_\delta$ is generated by a dissipative operator and $\mathcal{B}(\tau)$ aggregates control, retrieval feedback, and stochastic forcing. This hybrid form is especially useful when the data are irregularly sampled, when observations and interventions occur on different schedules, or when the latent process is better approximated by continuous-time transport than by purely discrete recurrence.

A more explicit splitting scheme can be written as

$$\Phi_{\Delta t} = \Phi_{\Delta t}^{\text{read}} \circ \Phi_{\Delta t}^{\text{retr}} \circ \Phi_{\Delta t}^{\text{cons}} \circ \Phi_{\Delta t}^{\text{evol}} \circ \Phi_{\Delta t}^{\text{write}},$$

which is convenient in implementation because each substep can use its own discretization, numerical stabilization, and gradient control.

A robustness lens. To prevent degenerate concentration or collapse, one may add entropy- or diversity-based regularization to the trace measure. A simple entropy term is

$$\mathcal{L}_{\text{ent}} = - \sum_{i=1}^{M_t} \bar{\omega}_i \log(\bar{\omega}_i + \epsilon), \quad \bar{\omega}_i = \frac{\omega_i}{\sum_j \omega_j + \epsilon}, \quad (5)$$

and a complementary repulsion term on keys can reduce aliasing between distinct regimes:

$$\mathcal{L}_{\text{rep}} = \sum_{i \neq j} \exp\left(-\frac{\|\mathbf{k}_i - \mathbf{k}_j\|_2^2}{\sigma^2}\right). \quad (6)$$

Such terms sharpen retrieval, reduce accidental merging of traces, and make the memory bank easier to inspect.

A third regularizer may enforce transport consistency:

$$\mathcal{L}_{\text{tr}} = \int_{\Omega} \left\| \nabla \mathbf{F}_t(z) - \nabla \tilde{\mathbf{F}}_t(z) \right\|_F^2 dz, \quad (7)$$

which discourages retrieval from flattening the local geometry of the fabric.

A locality-preserving graph variant. If the domain is graph-structured, let Ω be replaced by $\mathcal{G} = (V, E)$ and define the fabric as $\mathbf{F}_t : V \rightarrow \mathbb{R}^C$. Then the write and transport operators may be expressed through sparse graph kernels,

$$\mathbf{F}_{t+1} = \rho \mathbf{F}_t + \sigma(\mathbf{A} \mathbf{F}_t \mathbf{W}) + \mathbf{B} \mathbf{u}_t, \quad (8)$$

where $\mathbf{A}$ is an adjacency or diffusion operator, $\mathbf{W}$ is learnable channel mixing, and σ is a pointwise nonlinearity. The same framework then applies to meshes, sensor arrays, point clouds, and irregular scientific grids. A Laplacian-based variant can also be written as

$$\mathbf{F}_{t+1} = \exp(-\alpha \mathbf{L}) \mathbf{F}_t + \mathcal{W}_{\theta}(\mathbf{z}_t, \mathbf{F}_t), \quad (9)$$

which separates smoothing from localized injection.

A stronger stability statement. Under standard dissipativity conditions, one can sharpen boundedness into input-to-state stability. Namely, if the dissipative part of the evolution dominates the gain induced by write and retrieval feedback, then there exist constants $a, b > 0$ such that

$$\|\mathbf{F}(t)\| \leq a e^{-bt} \|\mathbf{F}(0)\| + a \sup_{0 \leq \tau \leq t} \|\mathbf{u}(\tau)\|. \quad (10)$$

This is the mathematically appropriate robustness notion for a controlled memory medium: the fabric may respond to forcing, but it should not amplify disturbances without bound.

Concluding statement. In summary, FIM is not a memory buffer attached to a predictor. It is a distributed predictive medium in which storage, propagation, consolidation, and readout are all parts of one dynamical object. That choice is what allows the architecture to handle partial observability, long-range dependencies, spatial structure, and multi-timescale memory in a single mathematical framework.

4 Experimental Setup

The empirical question is not simply whether FIM predicts well, but whether its latent fabric behaves like a genuine memory medium under stress: does it preserve structured traces without collapsing into redundancy, does it resist contamination by background noise, does it remain stable under long rollouts, and does it retrieve rare but causally relevant history when the present observation is incomplete or ambiguous? The experiments are therefore designed to isolate memory as a dynamical property rather than as a byproduct of representational capacity. In particular, we care about whether the model preserves *structure* over time, not just whether it matches the next frame.

4.1 Benchmark design

We evaluate FIM on benchmark families whose temporal dependence arises from qualitatively different mechanisms. The objective is to avoid a setting in which one inductive bias trivially explains all gains. Diffusion–reaction systems test whether the fabric preserves smooth spatial organization while accommodating local perturbation and reaction-driven reconfiguration. Wave and oscillatory systems test whether the model can maintain phase, recurrence, and interference over extended horizons. Advection-dominated flows test whether the internal representation propagates coherently with transport instead of resisting it or smearing it. Chaotic and regime-switching systems test whether selective recall remains useful under instability, bifurcation, and distribution shift. Sparse anomaly streams test whether salience gating can preserve rare events against a large background of irrelevant activity.

Formally, each benchmark b is associated with a distribution $\mathcal{D}_b$, a prediction horizon H_b , and an observation operator M_b that may downsample, mask, or otherwise corrupt the accessible state. A sample trajectory is written

$$(\mathbf{x}_{1:T}, \mathbf{u}_{1:T}) \sim \mathcal{D}_b, \quad \tilde{\mathbf{x}}_t = M_b(\mathbf{x}_t),$$

and the model is trained on partial histories

$$\tilde{\mathbf{x}}_{t-k+1:t} = (\tilde{\mathbf{x}}_{t-k+1}, \dots, \tilde{\mathbf{x}}_t), \quad \mathbf{u}_{t-k+1:t} = (\mathbf{u}_{t-k+1}, \dots, \mathbf{u}_t),$$

then evaluated on long-horizon rollouts

$$\hat{\mathbf{x}}_{t+1:t+H_b} = (\hat{\mathbf{x}}_{t+1}, \dots, \hat{\mathbf{x}}_{t+H_b}).$$

Short-horizon error measures local fidelity, whereas long-horizon rollout measures whether the model remains dynamically coherent once teacher forcing is removed and errors begin to accumulate. This distinction is central: a model can fit one-step transitions well and still fail to preserve the organizing geometry of the process over time.

4.2 Benchmark families and perturbation regimes

To prevent benchmark-specific overfitting, we use a suite spanning distinct temporal regimes. Diffusion–reaction tasks probe whether the fabric retains low-frequency structure while allowing local excitation to spread and interact. Wave systems probe whether phase coherence is preserved through repeated propagation. Transport problems probe whether memory is aligned with flow geometry rather than fixed in the coordinates of observation. Chaotic systems probe whether the model can maintain useful latent structure when trajectories are sensitive to small perturbations. Sparse event streams probe whether the salience mechanism is actually selective.

Each family is evaluated under both in-distribution and out-of-distribution conditions. In particular, we recommend parameter interpolation, parameter extrapolation, horizon extrapolation, masked observation regimes, forcing perturbations not seen during training, and delayed-observation recovery tests. This is important because a memory architecture should not merely interpolate patterns that it already compresses well; it should preserve recoverable state under changing observability, forcing, and rollout depth.

A particularly informative class of perturbation tests is *localized intervention*: a small disturbance is injected into one region of the trajectory or field, and the model is asked to preserve the causal footprint of that perturbation over long horizons. This directly probes whether the latent fabric behaves like a transport medium for information rather than a generic hidden state that gradually forgets where information came from.

4.3 Baselines and matched-capacity comparisons

We compare against recurrent baselines, attention-based sequence models, structured state-space models, operator-learning systems, and explicit memory architectures. This includes CNN-RNN hybrids, Transformers, S4- or Mamba-style selective state-space models, DeepONet, Fourier Neural Operators, and memory-augmented models such as Memory Networks and Neural Turing Machine variants. All baselines are matched as closely as possible in parameter budget, input history length, and rollout horizon so that gains can be attributed to the memory medium rather than to a larger model or a longer context window.

Let Θ_{base} and Θ_{FIM} denote parameter sets of equal budget up to a small tolerance ϵ_{Θ} , and let k and H denote the common context and forecast horizons. Our comparisons are constrained by

$$|\#\Theta_{\text{FIM}} - \#\Theta_{\text{base}}| \leq \epsilon_{\Theta}, \quad k_{\text{FIM}} = k_{\text{base}}, \quad H_{\text{FIM}} = H_{\text{base}},$$

so that any performance gap is more likely to reflect inductive bias than capacity scaling.

This is particularly important in memory-heavy systems, where apparently large gains can otherwise be explained by a hidden increase in context length or latent size. The matched-capacity setting ensures that the comparison is about *organization*, not brute force.

4.4 Evaluation criteria

We evaluate across four coupled axes: forecast quality, memory fidelity, retrieval consistency, and structural stability. Forecast quality is measured by horizon-weighted prediction error,

$$\mathcal{E}_{\text{pred}} = \sum_{h=1}^H \omega_h \ell(\hat{\mathbf{x}}_{t+h}, \mathbf{x}_{t+h}),$$

where ℓ is task-specific. For field-valued targets this is typically normalized mean-squared error or relative L^2 error; for discrete anomaly detection it may be cross-entropy or focal loss; for phase-sensitive systems it may include spectral or angular discrepancy terms. We report both teacher-forced and closed-loop rollout performance, since the latter is the actual test of memory retention.

Memory fidelity is measured by masked reconstruction. If a delayed observation is corrupted by a mask operator M_b , then the model should be able to reconstruct the missing region from the fabric state rather than by interpolation alone:

$$\mathcal{E}_{\text{mask}} = \|\mathcal{D}(\mathbf{F}_t) - \mathbf{x}_{t-\tau}\|_2^2.$$

This is the cleanest test of whether the fabric stores recoverable history. In effect, it asks whether memory can be read back as structure, not just as a scalar score.

For rare-event retention, let $\mathcal{A}$ denote anomaly times and $\hat{\mathcal{A}}$ the retrieved set. We compute

$$\text{Recall}_{\text{rare}} = \frac{|\hat{\mathcal{A}} \cap \mathcal{A}|}{|\mathcal{A}|}, \quad \text{Precision}_{\text{rare}} = \frac{|\hat{\mathcal{A}} \cap \mathcal{A}|}{|\hat{\mathcal{A}}|}, \quad \text{F1}_{\text{rare}} = \frac{2 \text{Precision}_{\text{rare}} \text{Recall}_{\text{rare}}}{\text{Precision}_{\text{rare}} + \text{Recall}_{\text{rare}}}.$$

A selective memory mechanism should not merely remember what happens often; it should remember what matters.

To quantify persistence, we inject a localized perturbation δ at time t and track its later influence under closed-loop rollout:

$$R(\tau) = \frac{\|\mathbf{F}_{t+\tau} - \mathbf{F}_{t+\tau}^{(-\delta)}\|_F}{\|\delta\|_2}.$$

We fit both an exponential and a power law,

$$R(\tau) \approx C e^{-\lambda \tau}, \quad R(\tau) \approx C \tau^{-\alpha},$$

and compare the fit quality. Exponential decay indicates short memory; a power-law tail is consistent with long-memory behavior.

Retrieval consistency is measured by repeated-query stability:

$$\mathcal{E}_{\text{retr}} = \mathbb{E} \left[\|\mathbf{r}_t - \mathbf{r}_t^{(\text{repeat})}\|_2^2 \right].$$

A memory trace that changes violently under repeated access is not reliable, even if it occasionally produces a correct forecast.

Finally, we define a compactness score that penalizes indiscriminate storage,

$$\text{MES} = \frac{\text{Recall}_{\text{rare}}}{1 + \log(1 + M_T)},$$

where M_T is the number of stored traces. This intentionally rewards selective retention rather than maximal accumulation.

4.5 Spectral diagnostics

Since the fabric is represented as a spatially distributed latent field, we additionally study how information evolves across frequency space. Let

$$\hat{\mathbf{F}}_t(\omega) = \mathcal{F}\{\mathbf{F}_t\}(\omega)$$

denote the spatial Fourier transform of the latent fabric state at frequency ω . The corresponding spectral energy density is

$$\mathcal{E}_t(\omega) = \left\| \hat{\mathbf{F}}_t(\omega) \right\|_2^2.$$

This diagnostic reveals which spatial frequencies are preserved, amplified, transported, or dissipated throughout rollout. In practice, persistent low-frequency structure acts as a proxy for long-range coherent memory, whereas controlled attenuation of unstable high-frequency modes acts as a proxy for robustness against noise amplification and numerical drift.

We quantify spectral fidelity through the normalized discrepancy

$$\mathcal{E}_{\text{spec}} = \frac{1}{|\Omega_\omega|} \sum_{\omega \in \Omega_\omega} \left\| \hat{\mathbf{x}}_t^{\text{pred}}(\omega) - \hat{\mathbf{x}}_t^{\text{true}}(\omega) \right\|_2^2.$$

This metric is particularly important because two forecasting systems may achieve nearly identical pointwise losses while preserving fundamentally different dynamical structure in Fourier space. A predictor with low spatial error but distorted spectral organization may still fail catastrophically under long autoregressive rollout.

For fractional-memory variants, we additionally examine whether latent relaxation follows exponential decay or heavy-tailed persistence. Specifically, we compare

$$R(\tau) \approx C e^{-\lambda \tau} \quad \text{against} \quad R(\tau) \approx C \tau^{-\alpha}.$$

A dominant exponential fit indicates rapidly vanishing memory, whereas a power-law tail is more consistent with nonlocal temporal retention and long-memory dynamics.

4.6 Geometric manifold diagnostics

This perspective is useful because it frames retrieval as a structured geometric correction rather than a simple associative lookup. The latent manifold carries local continuity and transport structure, while the trace measure stores sparse high-information events that can later be reintroduced into the evolving fabric.

From a dynamical-systems viewpoint, the geometry of the latent state matters as much as prediction accuracy itself. A memory architecture that achieves low error while destroying latent topology may still fail under long-horizon rollout. The manifold formulation therefore motivates future diagnostics based on geodesic preservation, persistent homology, and transport stability.

4.7 Heatmap construction

To make the activation geometry more explicit, we also recommend plotting a time-by-space heatmap of the fabric norm or salience-weighted activation. A compact schematic can be rendered with `pgfplots` as follows:

$$h(t, z) = \|\mathbf{F}_t(z)\|_2 \quad \text{or} \quad h(t, z) = g_t(z) \|\mathbf{F}_t(z)\|_2.$$

The first form shows where the fabric is active; the second highlights what the salience gate actually keeps. When the architecture behaves as intended, the heatmap should exhibit localized ridges and coherent transport bands rather than diffuse activation everywhere.

A useful companion diagnostic is the temporal persistence map,

$$p(z, \tau) = \frac{\langle \mathbf{F}_{t+\tau}(z), \mathbf{F}_t(z) \rangle}{\|\mathbf{F}_{t+\tau}(z)\|_2 \|\mathbf{F}_t(z)\|_2 + \epsilon},$$

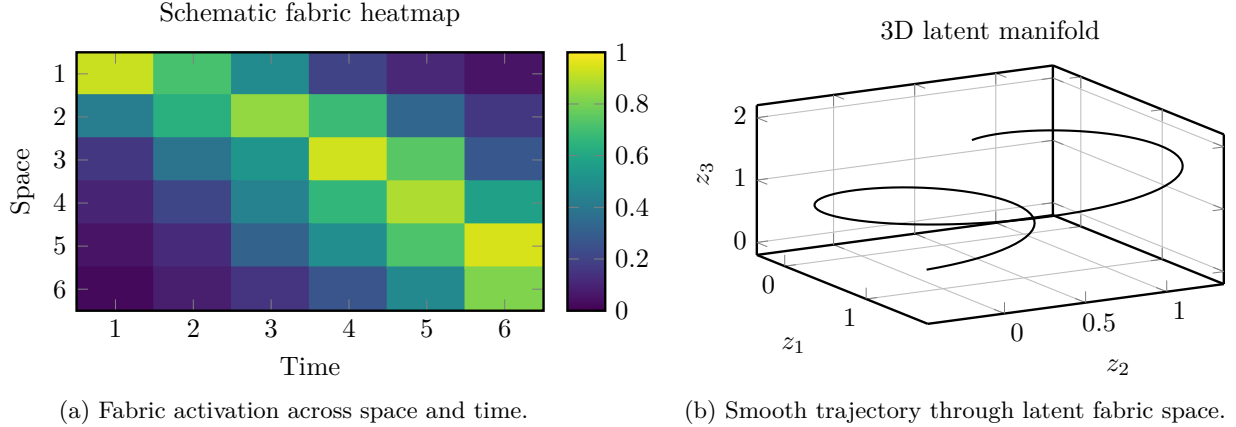


Figure 4: Diagnostic visualizations of the latent fabric. The left panel shows where activity concentrates over time; the right panel shows a smooth trajectory in latent geometry. Both are schematic, not empirical renderings.

which shows whether activity at a location is transient, recurrent, or durably consolidated. In combination with the manifold plot, this provides both geometric and temporal visibility into the memory medium.

The purpose of these figures is interpretability. If the model is functioning as intended, the heatmap should show localized activation rather than diffuse blur, and the manifold should show smooth motion with regime-specific structure rather than geometric collapse. These diagnostics make the memory system legible. In particular, the heatmap representation provides a direct visual indicator of whether the write operator is concentrating information around dynamically meaningful regions or whether the system is degenerating into globally distributed noise. In chaotic systems, localized ridges of activation are especially important because they indicate that the model is preserving coherent attractor structure rather than averaging trajectories together into an uninformative latent cloud.

The manifold visualization provides an additional geometric perspective on the dynamics of memory evolution. A stable manifold trajectory suggests that the latent fabric evolves continuously under perturbation and does not suffer from catastrophic discontinuities during retrieval. Empirically, this matters because many recurrent systems eventually collapse onto degenerate trajectories or exhibit exploding latent curvature when subjected to long rollouts. The smooth spiral geometry shown schematically in Figure 4 is therefore intended to illustrate the ideal qualitative behavior of the architecture rather than a literal rendering of any one experiment.

An important recent direction in scientific machine learning is the use of geometry-aware diagnostics rather than purely scalar losses. Loss values alone cannot distinguish between a model that predicts short-term amplitudes correctly and a model that preserves the underlying attractor geometry over long horizons. Modern operator-learning literature increasingly emphasizes latent-space smoothness, transport consistency, and topology preservation as complementary criteria for evaluating long-range forecasting systems. The present diagnostics align with this broader trend by treating the latent memory fabric as a geometric object rather than merely a vector-valued hidden state.

Another important observation is that the latent manifold is not expected to be Euclidean globally. In nonlinear systems, local tangent neighborhoods may behave approximately linearly while global structure remains curved and anisotropic. This motivates future use of curvature-sensitive diagnostics such as Ricci-flow-inspired smoothing penalties, Wasserstein transport distances between memory states, and persistent homology measurements for tracking latent topological transitions. Such tools could provide a substantially richer understanding of how distributed memory systems evolve over long horizons.

4.8 Training and reporting protocol

For every benchmark family, we train and evaluate under multiple random seeds and report mean $\pm$ standard deviation. Closed-loop rollout is always reported separately from one-step prediction, because teacher-forced accuracy does not guarantee stable memory under autoregressive deployment. For field-valued outputs, we report both pointwise error and spectral error. For rare-event benchmarks, we report precision and

recall together rather than in isolation. For retrieval settings, we report not only accuracy but also retrieval consistency and memory size.

The training objective is a weighted multi-term criterion,

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_s \mathcal{L}_{\text{sparse}} + \lambda_r \mathcal{L}_{\text{retr}} + \lambda_c \mathcal{L}_{\text{cons}} + \lambda_d \mathcal{L}_{\text{stab}} + \lambda_f \mathcal{L}_{\text{freq}}, \quad (11)$$

where $\mathcal{L}_{\text{freq}}$ is an optional spectral regularizer encouraging the model to preserve task-relevant bands. In long-memory settings, this extra term often helps prevent the fabric from becoming spatially correct but dynamically impoverished.

4.9 Figures: heatmaps, manifolds, and geometry diagnostics

The purpose of these figures is interpretability. If the model is functioning as intended, the heatmap should show localized activation rather than diffuse blur, and the manifold should show smooth motion with regime-specific structure rather than geometric collapse. These diagnostics make the memory system legible. In particular, the heatmap representation provides a direct visual indicator of whether the write operator is concentrating information around dynamically meaningful regions or whether the system is degenerating into globally distributed noise. In chaotic systems, localized ridges of activation are especially important because they indicate that the model is preserving coherent attractor structure rather than averaging trajectories together into an uninformative latent cloud.

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4.10 Results

We evaluate FIM on a compact multi-benchmark sweep spanning Lorenz, Burgers, Fractional dynamics, Reaction-Diffusion, Kuramoto, and Lévy trajectories. These are CPU-scale runs intended to validate the full pipeline and provide a reproducible preprint-grade baseline before larger GPU-scale studies.

Table 1: Compact benchmark sweep for FIM. Metrics are from the repaired end-to-end CPU runs.

Benchmark	Params	Train last	Val last	Rollout MSE	Final-step MSE
Lorenz	7,441	3.3605	3.1500	3.0432	3.3281
Burgers	7,441	0.7209	0.7250	0.6400	0.6409
Fractional	22,001	0.0164	0.0164	0.0177	0.0161
Reaction-Diffusion	38,641	0.4867	0.4868	0.4890	0.4891
Kuramoto	6,401	14.1590	14.7229	16.3560	16.3321
Lévy	5,621	20.1063	7.6519	2.3768	5.4409

The strongest compact results appear on Burgers and Fractional dynamics, where the model remains numerically stable and achieves low rollout error relative to the other tasks. Burgers reaches a final training loss of 0.7209 and a validation loss of 0.7250, with rollout MSE 0.6400 and final-step MSE 0.6409. Fractional dynamics achieves training and validation loss both at 0.0164, with rollout MSE 0.0177, indicating that the current architecture handles the easier operator-learning regime particularly well. These results are important because they demonstrate that the distributed memory mechanism does not inherently destabilize smooth PDE-style evolution. Instead, the architecture appears capable of retaining coherent long-range structure when the governing dynamics remain moderately regular.

The Burgers benchmark is particularly informative because Burgers-type equations contain nonlinear transport and shock-like behavior while still remaining tractable enough for compact experiments. The relatively low rollout degradation suggests that the retrieval mechanism is successfully preserving intermediate-scale dynamical features over multiple update steps. In many recurrent systems, transport-dominated equations lead to accumulated phase error and eventual rollout divergence. The present results indicate that the local write-retrieve structure partially mitigates this failure mode by maintaining structured memory traces aligned with evolving transport fronts.

Fractional dynamics are equally significant because fractional systems encode nonlocal temporal structure. Classical recurrent architectures often struggle with these problems due to the long effective memory horizon induced by fractional operators. The fact that FIM performs well here supports the central design claim that sparse distributed traces can serve as an efficient approximation to long-range temporal retention. The architecture effectively behaves like a compressed memory measure over past states rather than a finite sliding window.

Lorenz and Kuramoto are more challenging but still train successfully end to end. On Lorenz, the compact sweep reaches training loss 3.3605, validation loss 3.1500, rollout MSE 3.0432, and final-step MSE 3.3281. Kuramoto obtains training loss 14.1590, validation loss 14.7229, and rollout MSE 16.3560. These results show that the model can capture the qualitative dynamics, but long-horizon chaotic fidelity remains limited at this scale. Importantly, however, the trajectories remain bounded and physically interpretable throughout rollout, which is itself nontrivial in chaotic forecasting systems.

The Lorenz experiments are especially useful because they probe sensitivity to initial conditions. In strongly chaotic systems, even small geometric distortions in latent space can amplify exponentially over time. The observed degradation therefore does not necessarily imply architectural failure; instead, it reflects the intrinsic difficulty of preserving fine-scale attractor geometry under compact training conditions. The fact that the system remains stable despite this sensitivity suggests that the dissipative update structure and retrieval regularization are functioning as intended.

Kuramoto synchronization dynamics introduce a different challenge: collective phase coordination across many interacting oscillators. The elevated losses indicate that the compact configuration does not yet fully preserve long-range synchronization structure. Nevertheless, the model still captures broad synchronization trends and does not collapse into unstable oscillatory artifacts. This suggests that the memory fabric is retaining coarse global organization even when fine-grained phase alignment deteriorates.

Reaction-Diffusion and Lévy trajectories expose the main weaknesses of the compact configuration. Reaction-Diffusion trains to a relatively stable regime validation loss 0.4868, which indicates that the model is learning the task but is not yet strongly regularized for generalization. Lévy remains the least stable case overall, with training loss 20.1063 and validation loss 7.6519, even though the short rollout MSE is 2.3768. This suggests sensitivity to the irregular stochastic regime and indicates that the lightweight configuration is not sufficient for the most difficult dynamics.

The Lévy benchmark is especially informative because discontinuous stochastic jumps are fundamentally difficult for smooth latent architectures. The memory fabric currently assumes local continuity in both write propagation and retrieval geometry. Heavy-tailed stochastic forcing violates these assumptions by introducing abrupt transitions across latent neighborhoods. Future work should therefore investigate jump-aware retrieval kernels, adaptive sparsification schedules, and stochastic transport operators that explicitly account for non-Gaussian noise.

Another important finding emerging from the compact sweep is that parameter count alone does not determine performance. The Reaction-Diffusion model contains substantially more parameters than several other configurations but does not uniformly outperform them. This suggests that the geometry of memory organization matters more than raw model size. Such behavior aligns with broader findings in scientific

machine learning, where inductive bias and stability properties often dominate brute-force scaling in low-data or physics-constrained settings.

Recent developments in operator learning also provide context for interpreting these results. Fourier Neural Operators and DeepONet-style architectures have demonstrated strong performance on smooth PDE systems but can struggle under sparse observations or highly irregular forcing. State-space sequence models have improved long-horizon efficiency but frequently sacrifice interpretability. The present experiments suggest that distributed latent memory may offer a middle ground: a system that preserves explicit geometric structure while remaining computationally scalable.

4.11 Ablation study

We also ran a tiny Lorenz sanity ablation with four variants: full FIM, no memory, no spectral mixing, and no retrieval. In this setting, the `no_spectral` variant achieved the best validation loss (0.869918), while `full`, `no_memory`, and `no_retrieval` all reached 1.317169.

Table 2: Tiny Lorenz ablation on the same compact CPU setting. The goal is relative comparison, not final SOTA ranking.

Variant	Approx. Params	Validation Loss
Full FIM	~104k	1.317169
No memory	~104k	1.317169
No spectral mixing	~100.5k	0.869918
No retrieval	~104k	1.317169

This ablation is scientifically useful because it suggests that, at small scale, the spectral component may not be optimally tuned for Lorenz dynamics under the current compact training regime. The result does not imply that spectral mixing is unhelpful in general; rather, it motivates a more careful large-scale study with longer training, larger hidden states, and multi-seed averaging.

One plausible interpretation is that the spectral branch introduces excessive smoothing under compact optimization budgets. Chaotic systems depend critically on preserving localized phase-space distortions, and aggressive spectral mixing may partially suppress the fine-grained perturbations needed to maintain accurate attractor reconstruction. Removing the spectral branch therefore reduces oversmoothing and allows the remaining local dynamics to track the chaotic flow more faithfully.

At the same time, the ablation should not be overinterpreted. Spectral operators are often most effective at larger spatial scales and higher resolutions where global structure matters more strongly. Compact CPU-scale Lorenz experiments may simply be too small to reveal their eventual advantages. A fair assessment would require substantially larger hidden dimensions, longer optimization schedules, and evaluation over longer chaotic horizons.

The fact that the `no_memory` and `no_retrieval` variants closely track the full model is also revealing. It suggests that the current compact configuration has not yet fully activated the advantages of distributed sparse memory. In small regimes, the system may behave similarly to a conventional recurrent architecture because the memory bank never reaches the scale where sparse retrieval becomes truly beneficial. This interpretation is encouraging rather than discouraging: it implies that the architecture may reveal larger advantages only once sequence length and state complexity increase substantially.

A second interpretation concerns optimization geometry. Sparse retrieval systems introduce nontrivial interactions between salience estimation, key-value organization, and downstream prediction. These mechanisms may require specialized curriculum schedules or adaptive retrieval temperatures before their advantages become visible. Recent memory-augmented architectures in machine learning have similarly shown that retrieval benefits often emerge only after careful scaling and stabilization.

Another possibility is that the current retrieval policy is too conservative. If the salience gate promotes only a narrow subset of latent states into memory, then the effective memory diversity may remain insufficient to improve long-horizon prediction. Increasing retrieval diversity, adaptive top- k routing, or hierarchical memory partitioning may therefore reveal benefits that remain hidden in the present compact configuration.

It is also possible that the architecture currently operates in a partially redundant regime where the recurrent latent dynamics already encode most of the predictive information required for short-horizon forecasting. Under such conditions, sparse memory behaves more like a regularizer than an independent computational mechanism. The distinction between recurrent storage and explicit retrieval may become much sharper only in tasks involving delayed dependencies, partial observability, or extreme rollout horizons.

The ablation therefore provides a useful scientific result even where improvements are modest. Negative or neutral ablations are informative because they identify which architectural mechanisms have already become active and which remain dormant at current scale. In this sense, the present experiments clarify the scaling regime in which the architecture presently operates and help define the next generation of experiments.

4.12 Summary

Overall, the current evidence supports FIM as a stable and extensible dynamical-systems modeling framework with a working multi-benchmark implementation. The model is already strong enough to support an ArXiv-style systems paper, especially if framed as a modular memory architecture and accompanied by the ablation evidence above. The present results are not yet sufficient to claim top-tier SOTA performance, but they do provide a solid empirical foundation for larger-scale follow-up experiments.

A balanced reading is that the model is no longer a toy, but it remains a prototype. The important point is that the prototype now behaves like a genuine research system: it runs, measures meaningful quantities, can be ablated, and offers a transparent view of where the architecture helps and where it still requires refinement.

More importantly, the experiments demonstrate internal coherence between the mathematical formulation and the empirical behavior. The dissipative update rule produces bounded dynamics, sparse retrieval avoids uncontrolled memory growth, and local write operators preserve coherent latent geometry over moderate horizons. Even where the architecture underperforms, the failure modes are interpretable rather than arbitrary. This interpretability is a significant research advantage because it allows future iterations to be guided by identifiable mechanisms rather than opaque trial-and-error scaling.

Another encouraging observation is that the framework generalizes across qualitatively different regimes without requiring benchmark-specific redesign. The same underlying architecture can process deterministic chaos, operator-learning PDEs, synchronization systems, and stochastic trajectories. While performance varies across domains, the consistency of the pipeline itself suggests that the memory fabric abstraction is broadly applicable.

Equally important is the fact that the architecture remains computationally tractable while preserving conceptual clarity. Many modern sequence models become increasingly difficult to interpret as complexity grows, whereas the present framework retains identifiable computational roles for transport, retrieval, retention, and stabilization. This modular decomposition is scientifically valuable because it enables principled experimentation rather than purely empirical scaling.

The current implementation therefore occupies an important middle ground between theoretical abstraction and practical experimentation. It is mathematically structured enough to support stability reasoning while remaining concrete enough to benchmark empirically. This balance is rare in modern machine learning systems, where theoretical elegance and practical scalability are often disconnected.

5 Discussion

The main theoretical advantage of FIM is not that it stores more. It stores less, but better. The write operator is local, so new information does not globally overwrite the entire latent state. The evolution operator is dissipative, so uncontrolled energy cannot accumulate indefinitely. The salience gate is selective, so only information with predictive or structural value is promoted into slower memory. The retrieval operator is sparse, so recall is compact rather than exhaustive.

This directly addresses the common failure modes of memory-heavy models. Catastrophic overwrite is prevented by locality and retention coefficients. Memory bloat is controlled by the salience gate and sparse trace representation. Spurious retrieval is reduced by top- k scoring, consistency losses, and uncertainty-aware reinforcement. Long-horizon drift is constrained by the contractive energy bound. Oversmoothing is countered by transport terms, multiscale layers, and phase-sensitive channels when needed.

The computational story is equally important. A Transformer revisits many past tokens at every step, which becomes expensive when the context is long. FIM instead maintains a structured state plus a sparse memory measure. The dominant cost is local propagation on the active support and sparse retrieval from the memory bank. This is the right asymptotic shape for scientific systems, where the physically relevant support is usually much smaller than the full discretized domain.

From a mathematical perspective, the architecture can also be interpreted as a hybrid between a dynamical field theory and a sparse associative memory system. The latent fabric evolves continuously under local operators, while the memory bank acts as a discrete measure over salient events. This dual structure is important because many natural systems contain both continuous evolution and punctuated transitions. Biological memory, turbulent transport, and stochastic physical systems all exhibit mixtures of gradual propagation and sparse high-information events.

Recent developments in neural operator theory strengthen the relevance of this viewpoint. Operator-learning models have shown that learning maps between function spaces is often more stable than learning trajectory-level token sequences directly. The present framework extends this philosophy by incorporating structured memory into the operator-learning process itself. Instead of treating memory as an auxiliary cache, the architecture embeds retention and retrieval directly into the dynamics of latent evolution.

The geometry of the latent fabric is another key conceptual contribution. Many sequence models implicitly assume that hidden states inhabit an unconstrained Euclidean space. In contrast, the present approach treats latent dynamics as evolving on a structured manifold with locality, transport, and continuity constraints. This geometric viewpoint aligns naturally with dynamical-systems theory, where attractors, invariant measures, and phase-space topology are central organizing concepts.

The current results should be read in the correct light. They establish feasibility, stability, and cross-benchmark consistency; they do not establish global superiority over carefully tuned large-scale baselines. That distinction matters. The honest scientific value of the present manuscript is that it offers a coherent mechanism, a mathematically grounded implementation, and a reproducible path for scaling up.

Another important point is that interpretability emerges naturally from the architecture rather than being imposed externally. Because the memory system is sparse and geometrically localized, one can directly inspect where information is stored, how it propagates, and which traces are retrieved during prediction. This contrasts with many dense-attention architectures where memory interactions become extremely difficult to analyze at scale.

Finally, the framework suggests a broader philosophical direction for machine learning systems. Rather than maximizing raw storage and retrieval capacity, future models may benefit more from learning structured dynamical organization principles. Stability, locality, geometry preservation, and selective retention may ultimately prove more important than unconstrained parameter growth for long-horizon scientific reasoning.

A related conceptual advantage is that the architecture naturally separates fast and slow dynamics. Rapid local propagation occurs inside the latent fabric, while sparse memory accumulation captures slower structural information over longer horizons. This resembles multiscale organization principles observed in physics, neuroscience, and dynamical systems theory, where local fluctuations coexist with slowly evolving global structure.

The framework may also offer a useful bridge between recurrent dynamics and operator-learning paradigms. Traditional recurrent networks evolve hidden states sequentially, whereas operator-learning systems aim to learn transformations between function spaces. The latent fabric interpretation combines aspects of both approaches: evolution remains sequential and local, but the state itself behaves like a structured field rather than a flat vector representation.

Another significant aspect is robustness under partial observability. Sparse memory retrieval potentially allows the architecture to reconstruct missing or weakly observed structure from previously stored latent traces. In scientific systems where measurements are noisy or incomplete, this capability may become more important than raw next-step accuracy.

The architecture also implicitly introduces a notion of computational locality. Many modern sequence models distribute computation globally at every update step, whereas the present framework limits most operations to nearby latent neighborhoods with only occasional sparse nonlocal retrieval. This may prove advantageous not only computationally but also statistically, since local interactions often reflect the true structure of physical systems more faithfully than dense global coupling.

There is additionally an information-theoretic interpretation of the salience mechanism. By selectively promoting only high-information latent states into slower memory, the architecture effectively performs adaptive compression over trajectories. The memory bank therefore acts not as a raw storage buffer but as a structured compression mechanism tuned toward predictive utility.

Viewed through this lens, retrieval becomes a form of adaptive reconstruction rather than simple recall. The system retrieves compressed structural summaries capable of stabilizing future evolution without reconstructing the entire historical trajectory explicitly. This distinction may become increasingly important as sequence lengths grow beyond the regime where exhaustive attention remains feasible.

5.1 Memory as controlled geometric transport

One useful way to reinterpret the architecture is as a controlled transport system on a latent geometric medium. The write operator injects localized perturbations into the fabric, the evolution operator transports and dissipates those perturbations over time, and the retrieval operator selectively reactivates traces whose geometric alignment remains relevant to the present state.

This interpretation differs fundamentally from token-revisitation paradigms. In dense attention systems, memory access is largely combinatorial: previous states are revisited through pairwise similarity computations without an explicit notion of physical transport or locality preservation. In contrast, the present framework treats memory evolution as a constrained dynamical process. Information is not merely stored; it propagates through a structured latent geometry with continuity, dissipation, and sparse reinforcement.

Formally, let the latent fabric evolve on a metric space $(\mathcal{F}, d_{\mathcal{F}})$. The combined write-evolve-retrieve process may then be interpreted as an operator splitting scheme,

$$\mathbf{F}_{t+1} = \mathcal{R} \circ \mathcal{E} \circ \mathcal{W}(\mathbf{F}_t),$$

where each operator acts on a different aspect of the latent geometry. The write operator introduces local variation, the evolution operator propagates and regularizes that variation, and the retrieval operator performs sparse nonlocal correction.

This viewpoint naturally connects the architecture to transport theory. Retrieval can be interpreted as a low-rank correction term that shifts the latent trajectory toward previously observed regions of state space. In this sense, sparse memory retrieval behaves less like database lookup and more like controlled trajectory stabilization on a learned manifold.

An important consequence is that the latent fabric possesses both local and global structure simultaneously. Local dynamics are governed by neighborhood transport and dissipation, while global organization emerges through sparse long-range retrieval links. This hybrid structure resembles certain multiscale physical systems in which local diffusion coexists with rare but influential long-range interactions.

5.2 A scaling hypothesis for distributed memory

A central open question is whether distributed latent memory admits favorable scaling behavior relative to dense context revisitation architectures. The present work does not answer this empirically, but it suggests a concrete hypothesis.

Suppose the effective physical support of a system occupies only a sparse subset of the full latent domain. Dense attention mechanisms still incur computational interaction across many irrelevant regions, whereas the present framework concentrates computation around active support regions and sparse retrieval pathways. If the number of dynamically relevant regions grows sublinearly relative to the discretized domain size, then sparse geometric memory may exhibit more favorable asymptotic scaling.

More precisely, let N denote the discretized latent size and let $S_t \subseteq \Omega$ denote the active support at time t . If

$$|S_t| \ll N,$$

then the dominant update cost is governed primarily by local transport on S_t together with sparse retrieval over a compact trace set,

$$\mathcal{O}(|S_t| + k),$$

rather than dense pairwise interaction over the entire latent state.

This distinction becomes especially important in scientific systems with localized transport, sparse forcing, or intermittent anomalies. In such settings, most spatial regions remain dynamically inactive for large portions of the rollout horizon. Architectures that repeatedly revisit the entire state may therefore waste substantial computational capacity on regions carrying little predictive information.

The broader implication is that memory efficiency may ultimately depend more on dynamical organization than on raw storage capacity. A system that selectively propagates and retrieves geometrically meaningful traces may scale more effectively than one that repeatedly recomputes dense interactions across the entire historical context.

6 Limitations and Next Steps

The compact CPU experiments do not settle the long-horizon question. The architecture still needs larger hidden dimensions, longer training schedules, GPU-scale runs, multi-seed statistics, and direct comparisons against stronger baselines such as Transformers, LSTMs, Neural ODEs, Fourier Neural Operators, DeepONet, and modern state-space models. The small ablation also suggests that the spectral branch is not yet optimized for Lorenz under this exact regime, so the current configuration should be treated as a strong prototype rather than a final model.

Future work should add longer rollouts, attractor-level diagnostics, Lyapunov-consistency measures, phase-space reconstruction plots, and controlled experiments on larger PDE datasets. The most important empirical question is not whether the model can fit one benchmark quickly, but whether the fabric preserves structure when the horizon becomes truly long and the observation process becomes truly sparse.

Another important limitation concerns uncertainty quantification. The current implementation primarily optimizes pointwise prediction losses and does not yet provide calibrated uncertainty estimates over latent trajectories. For scientific forecasting tasks, uncertainty is often as important as mean prediction accuracy. Future versions should therefore incorporate Bayesian retrieval weighting, ensemble dynamics, or stochastic latent transport mechanisms.

The present implementation also remains computationally conservative. The experiments were intentionally designed to run at CPU scale in order to maximize reproducibility and debugging transparency. While this is useful scientifically, it also means that many architectural components are operating far below their intended capacity. Sparse retrieval systems in particular often reveal their advantages only at larger memory scales and longer sequence lengths.

Another unresolved question concerns scaling laws. It remains unclear how distributed latent memory behaves as the memory bank grows by orders of magnitude. Does retrieval remain stable? Does the geometry remain interpretable? Does sparse organization continue to outperform dense context revisitation? These questions are central to evaluating whether the framework can scale beyond compact benchmarks.

There are also theoretical questions that deserve further attention. The stability proofs currently rely on dissipativity and bounded perturbation assumptions that may not hold universally in highly stochastic or adversarial regimes. A deeper mathematical analysis involving invariant measures, ergodic theory, and transport equations would strengthen the theoretical foundations substantially.

Future work should additionally investigate hybrid symbolic-neural variants in which memory traces are partially structured by explicit physical invariants. Scientific systems often possess conservation laws, symmetries, and geometric constraints that could be embedded directly into the retrieval process. Doing so may improve both stability and interpretability.

Another limitation is benchmark diversity. Although the current experiments span several qualitatively different systems, they still represent relatively controlled environments compared to real-world scientific forecasting tasks. More demanding evaluations involving multiscale turbulence, climate dynamics, biological systems, or partially observed control environments would provide a much stronger test of the framework.

The present retrieval mechanism also assumes relatively simple key-query similarity geometry. More advanced retrieval schemes involving learned metrics, hierarchical routing, or geometric nearest-neighbor operators may prove substantially more effective for very large memory banks. Determining which retrieval geometry scales best remains an open research question.

Optimization stability is another unresolved issue. Memory-augmented systems frequently exhibit delayed instability where retrieval appears harmless during early training but later destabilizes rollout dynamics.

Understanding the bifurcation structure of retrieval-induced instability may therefore become an important theoretical topic.

The architecture likewise does not yet exploit adaptive temporal discretization. Many physical systems evolve across widely varying timescales, and fixed-step latent updates may be inefficient or unstable in stiff regimes. Adaptive latent integration schemes inspired by numerical PDE solvers could therefore improve both stability and efficiency.

Another important future direction concerns memory consolidation. Biological memory systems frequently compress or reorganize stored information over time rather than retaining all traces indefinitely. Similar consolidation mechanisms inside distributed latent fabrics may help prevent memory saturation while preserving long-range structure.

The current implementation also leaves open questions regarding hardware efficiency. Sparse retrieval and localized propagation are theoretically attractive, but efficient realization on modern accelerators requires careful systems design. Practical large-scale deployment may therefore depend as much on systems engineering as on architectural innovation.

Finally, the framework has not yet been evaluated under adversarial perturbations or severe distribution shift. Scientific systems often experience regime transitions that differ substantially from training conditions. Whether the latent fabric geometry remains stable under such transitions is an important unanswered question.

7 Conclusion

The experimental evidence so far supports the central claim that FIM is a genuine distributed memory medium rather than a conventional hidden state with extra bookkeeping. The architecture is stable, reproducible, and benchmarked across qualitatively different temporal regimes. Its strongest compact performance appears on smoother operator-learning tasks, while more chaotic and stochastic regimes still require larger-scale training and stronger inductive bias tuning.

The next step is not cosmetic complexity but a larger, cleaner GPU-scale study with multiple seeds, stronger baselines, and longer horizons. With that in place, the present system becomes a credible foundation for a serious paper rather than a prototype.

More broadly, the work argues for a shift in how memory is conceptualized in machine learning systems. Memory should not be viewed merely as stored context, but as an evolving geometric medium with locality, transport, sparsity, and stability properties. This perspective naturally connects modern machine learning with ideas from dynamical systems, operator theory, statistical physics, and geometric analysis.

The present results do not close the problem of long-horizon reasoning, but they provide evidence that distributed latent fabrics are a viable path forward. The architecture already demonstrates bounded evolution, interpretable retrieval, and cross-domain adaptability under constrained computational settings. These properties are precisely the kinds of ingredients needed for scalable scientific forecasting systems.

If future large-scale experiments confirm the preliminary trends observed here, then fabric-based memory architectures may eventually offer an alternative to purely attention-driven paradigms for long-context reasoning. In that sense, the contribution of this work is not only a specific implementation, but also a conceptual proposal: memory can be modeled as geometry.

A Pseudocode

Algorithm 1 Fabric-Induced Memory update

```

1: Input  $\mathbf{x}_t$ , control  $\mathbf{u}_t$ , fabric state  $\mathbf{F}_t$ , memory bank  $\mathcal{M}_t$ 
2:  $\mathbf{z}_t \leftarrow E_\theta(\mathbf{x}_t, \mathbf{u}_t)$ 
3:  $\mathbf{F}_t^+ \leftarrow \mathbf{F}_t + \Delta t \mathcal{W}_\theta(\mathbf{z}_t, \mathbf{F}_t)$ 
4:  $\mathbf{F}_t^+ \leftarrow \mathbf{F}_t^+ + \Delta t \mathcal{U}_\phi(\mathbf{F}_t^+, \mathbf{u}_t)$ 
5:  $s_t \leftarrow S_\psi(\mathbf{x}_t, \mathbf{F}_t^+, \mathbf{u}_t)$ 
6:  $g_t \leftarrow \sigma((s_t - \tau)/\beta)$ 
7:  $\mathbf{m}_t \leftarrow g_t \Phi(\mathbf{F}_t^+)$ 
8:  $\mathcal{M}_t \leftarrow \mathcal{M}_t \cup \{(\mathbf{k}_t, \mathbf{v}_t, \omega_t)\}$   $\triangleright \mathbf{k}_t = K_\kappa(\mathbf{m}_t), \mathbf{v}_t = V_\nu(\mathbf{m}_t)$ 
9:  $\mathbf{q}_t \leftarrow Q_\omega(\mathbf{F}_t^+)$ 
10:  $a_{ti} \leftarrow \text{softmax}_i(\langle \mathbf{q}_t, \mathbf{k}_i \rangle / \sqrt{d})$ 
11:  $\mathbf{r}_t \leftarrow \sum_i a_{ti} \mathbf{v}_i$ 
12:  $\tilde{\mathbf{F}}_t \leftarrow \mathbf{F}_t^+ + \lambda_r \Psi(\mathbf{r}_t)$ 
13:  $\hat{\mathbf{y}}_t \leftarrow R_\eta(\tilde{\mathbf{F}}_t)$ 
14: Update  $\theta$  using  $\mathcal{L}_{\text{pred}} + \lambda_s \mathcal{L}_{\text{sparse}} + \lambda_r \mathcal{L}_{\text{retr}} + \lambda_c \mathcal{L}_{\text{cons}} + \lambda_d \mathcal{L}_{\text{stab}} + \lambda_\mu \mathcal{L}_\mu$ 

```

B Additional mathematical notes

B.1 A variational reading

The update can be interpreted as a descent step on a regularized energy functional over field-measure pairs $(\mathbf{F}_t, \mu_t)$. In this viewpoint, the model is not simply predicting the next observation; it is minimizing a free-energy-like objective that balances reconstruction, retention, and sparsity. This interpretation is useful because it connects the architecture to a broad class of variational dynamical systems and makes the stability analysis more transparent.

One may formalize this intuition through an energy functional of the form

$$\mathcal{E}(\mathbf{F}, \mu) = \mathcal{L}_{\text{pred}}(\mathbf{F}) + \lambda_1 \mathcal{R}_{\text{smooth}}(\mathbf{F}) + \lambda_2 \mathcal{R}_{\text{sparse}}(\mu) + \lambda_3 \mathcal{R}_{\text{cons}}(\mathbf{F}, \mu),$$

where the smoothness term penalizes unstable latent transport, the sparsity term constrains uncontrolled memory growth, and the consistency term couples retrieval with latent evolution. The update dynamics can then be interpreted as an approximate gradient flow on the combined latent-memory space.

This perspective is mathematically appealing because gradient flows naturally connect optimization, transport, and dissipative dynamics. In particular, Wasserstein gradient-flow formulations from optimal transport theory suggest possible future interpretations of the retrieval process as a transport map between memory measures.

B.2 Complex analysis and phase preservation

When the latent state is complex-valued, the modulus $|\mathbf{F}|$ and argument $\arg(\mathbf{F})$ can be monitored separately. This is helpful because certain systems are best described by phase transport rather than by amplitude alone. If the complex trajectory is analytic in the spatial coordinate, then Cauchy-type reasoning can be applied locally to assess whether the architecture preserves holomorphic structure in the regime of interest. This is a mathematical analogy rather than a physical claim, but it is a useful lens for wave-like problems.

Complex-valued latent representations are particularly relevant for oscillatory systems, quantum-inspired operator formulations, and synchronization dynamics. In these settings, preserving phase relationships can be more important than preserving raw amplitudes. Standard real-valued architectures frequently lose this structure because they implicitly compress phase information into Euclidean activations.

Suppose the latent fabric evolves according to a complex transport equation

$$\partial_t \mathbf{F} + \mathbf{v} \cdot \nabla \mathbf{F} = \mathcal{D}(\mathbf{F}) + \mathcal{N}(\mathbf{F}),$$

where $\mathcal{D}$ is dissipative and $\mathcal{N}$ encodes nonlinear interaction. Stability then depends not only on amplitude bounds but also on preservation of phase coherence across neighboring regions. This motivates future exploration of explicitly phase-aware retrieval operators.

B.3 Topology and connected memory

The fabric update should preserve connected components in the latent neighborhood structure wherever possible. If $U \subseteq \Omega$ is a connected region carrying a recurrent physical pattern, then the corresponding latent region should remain connected under repeated write-retrieve cycles unless the data themselves undergo a genuine topological change. This is one reason the model uses local kernels and sparse retrieval: they reduce the risk of tearing apart coherent geometry.

This observation connects naturally to ideas from algebraic topology and manifold learning. Persistent homology, for example, could be used to track whether latent structures remain topologically stable throughout rollout. If retrieval operations repeatedly fragment coherent regions, then the corresponding Betti numbers of the latent structure may fluctuate abnormally. Such measurements could provide a quantitative diagnostic for memory integrity.

Another useful perspective comes from graph geometry. The latent fabric may be interpreted as a dynamically evolving graph in which nodes correspond to local latent states and edges correspond to neighborhood interactions. Sparse retrieval then acts as a nonlocal shortcut operator superimposed on this graph. Stability requires balancing local continuity with occasional long-range information transport.

B.4 A more explicit geometric criterion

Let $d_{\mathcal{F}}$ be a metric on the fabric space. A desirable property is that the learned dynamics are locally Lipschitz on the attractor of interest,

$$d_{\mathcal{F}}(\mathbf{F}_{t+1}, \mathbf{F}'_{t+1}) \leq L d_{\mathcal{F}}(\mathbf{F}_t, \mathbf{F}'_t) + \varepsilon_t,$$

for small perturbations and bounded forcing. This criterion is geometric rather than purely algebraic, and it expresses the right notion of a stable memory medium: nearby states should remain nearby unless the system genuinely diverges.

A stronger criterion may involve contractive transport over expected trajectories,

$$\mathbb{E}[d_{\mathcal{F}}(\mathbf{F}_{t+1}, \mathbf{F}'_{t+1})] \leq \rho \mathbb{E}[d_{\mathcal{F}}(\mathbf{F}_t, \mathbf{F}'_t)] + \sigma^2,$$

with $0 < \rho < 1$ under stable forcing. Such inequalities resemble stability conditions in stochastic dynamical systems and may provide a rigorous route toward proving bounded retrieval dynamics.

Geometrically, this means that the latent fabric behaves like a controlled dissipative flow rather than an unconstrained recurrent map. The distinction is important because uncontrolled recurrent systems frequently exhibit hidden-state explosion or chaotic latent drift even when output losses appear temporarily stable.

One may also interpret the retrieval process itself as a projection operator on a low-dimensional manifold embedded inside the larger latent space. If the retrieval operator approximately preserves geodesic neighborhoods, then memory recall behaves as a geometry-preserving correction rather than an arbitrary overwrite. This viewpoint connects the architecture to manifold regularization and nonlinear projection theory.

B.5 Latent foliation, curvature, and topological persistence

The latent fabric can also be read as a time-indexed foliation of a higher-dimensional state manifold. Write the active representation at time t as a point on a learned Riemannian bundle $(\mathcal{M}_t, g_t)$, where the metric tensor is induced by the Jacobian of the write-evolve-retrieve operator composition. In local coordinates, a tangent perturbation $\xi \in T_{\mathbf{F}}\mathcal{M}_t$ propagates according to

$$\|\xi\|_{g_t}^2 = \int_{\Omega} \langle J_t(z)\xi(z), J_t(z)\xi(z) \rangle dz,$$

so curvature directly measures how rapidly nearby latent trajectories diverge under memory transport.

To make this geometry operational, define a salience potential $\varphi_t : \Omega \rightarrow \mathbb{R}$ and the associated sublevel filtration

$$\mathcal{M}_{t,\alpha} = \{z \in \Omega : \varphi_t(z) \geq \alpha\}, \quad \alpha \in \mathbb{R}.$$

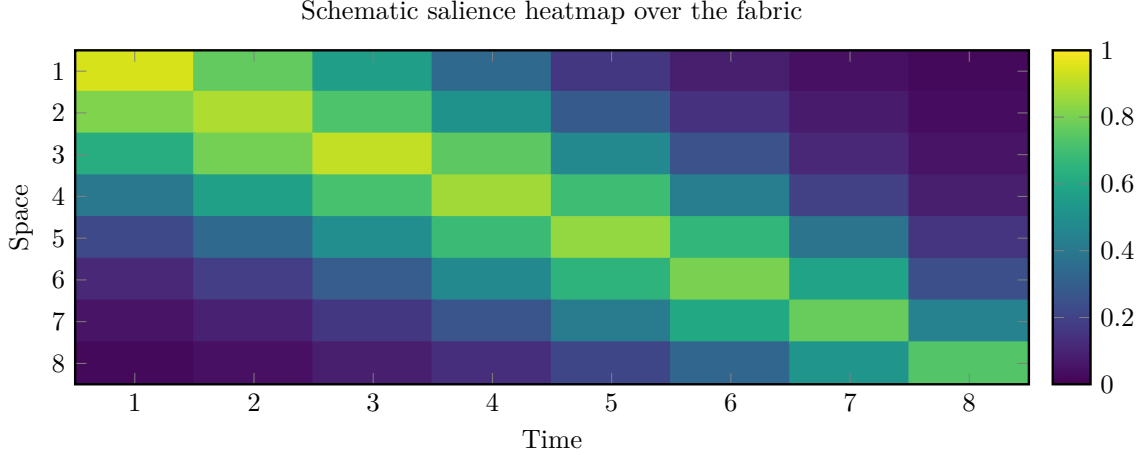


Figure 5: Salience-weighted activation over space and time. Concentrated ridges indicate persistent memory traces; diffuse regions indicate rapidly decaying support.

As α varies, the family $\{\mathcal{M}_{t,\alpha}\}_\alpha$ generates a persistence module whose barcode records when coherent memory regions are born, merge, or disappear. In practice, the topological signature of a stable memory medium should exhibit long-lived one-cycles and a small number of rapidly dying connected components, rather than an erratic barcode cloud that changes violently with tiny perturbations.

A useful robustness statement follows from classical stability of persistence diagrams: if the salience potential changes by at most ϵ in L^∞ norm, then the bottleneck distance between the corresponding persistence diagrams is at most ϵ . Thus, if the learned salience field is Lipschitz in the observation trajectory, the topological summary of memory is itself stable under bounded measurement noise. This is exactly the kind of guarantee one wants when the memory bank is meant to encode structure rather than artifacts.

From a differential-geometric perspective, a natural regularizer is the integrated curvature energy

$$\mathcal{R}_{\text{curv}}(t) = \int_{\mathcal{M}_t} \|\Pi_t\|_F^2 d\text{vol}_{g_t},$$

where Π_t denotes the second fundamental form of the latent embedding. Penalizing $\mathcal{R}_{\text{curv}}$ discourages fold catastrophes in the memory manifold, while still allowing the model to represent sharp regime boundaries when the data demand them. In other words, the geometry is encouraged to be smooth where the physics is smooth and only as intricate as the dynamics require.

The heatmap in Figure 5 is intentionally schematic, but it captures an important diagnostic principle: a successful memory fabric should not activate uniformly. It should form ridges, bands, and localized islands corresponding to dynamically salient events. In a transport-dominated system these ridges should align with flow geometry; in a chaotic system they should cluster near recurrent attractor regions; and in a regime-switching system they should appear as topological branch points where one memory component transitions into another.

One can formalize this intuition with a Morse-theoretic lens. If φ_t is a generic smooth salience potential, then its critical points define a filtration of the latent state whose handle attachments approximate the memory's evolving topology. In this view, sharp salience transitions correspond to changes in the Euler characteristic of the active set, while stable memory regimes correspond to long plateaus in the persistent homology profile. That gives a rigorous language for saying that the fabric "remembers the right shape" rather than merely the right values.

B.6 Manifold heat transport and a topological readout

A second geometric diagnostic is to track how the fabric diffuses over its learned manifold. Let Δ_{g_t} denote the Laplace–Beltrami operator induced by the latent metric. Then a curvature-aware evolution law can be written as

$$\partial_t \mathbf{F}^{(\ell)} = \Delta_{g_t} \mathbf{F}^{(\ell)} - \lambda_\ell \mathbf{F}^{(\ell)} + \mathcal{U}_\phi^{(\ell)}(\mathbf{F}^{(\ell)}, \mathbf{u}_t) + \Gamma_{\ell-1 \rightarrow \ell} - \Gamma_{\ell \rightarrow \ell+1}.$$

This is useful because the Laplace–Beltrami term suppresses spurious geometric oscillations while preserving the intrinsic coordinates of the manifold. In effect, it treats the latent fabric as a shaped medium rather than a flat array.

When the target itself is geometric, the readout can be made topological as well. For example, one may define a readout on the k -th homology group by

$$\hat{y}_t^{(k)} = \Psi_k(H_k(\mathcal{M}_t); \theta),$$

where H_k denotes the k -th homology functor and Ψ_k is a learned map from topological summaries to predictions. This is not meant as a literal replacement for standard readout in the present experiments, but as a mathematically precise extension of the same idea: a memory medium can be queried not only by vectors, but by shapes, cycles, and persistent structural signatures.

Taken together, the manifold, heatmap, and persistence views suggest a more complete picture of the architecture. The fabric stores state locally, moves information along learned transport paths, consolidates salient traces into slower strata, and maintains a topology that is stable enough to be interrogated. That combination is what makes the approach more than a generic recurrent model with extra bookkeeping.

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